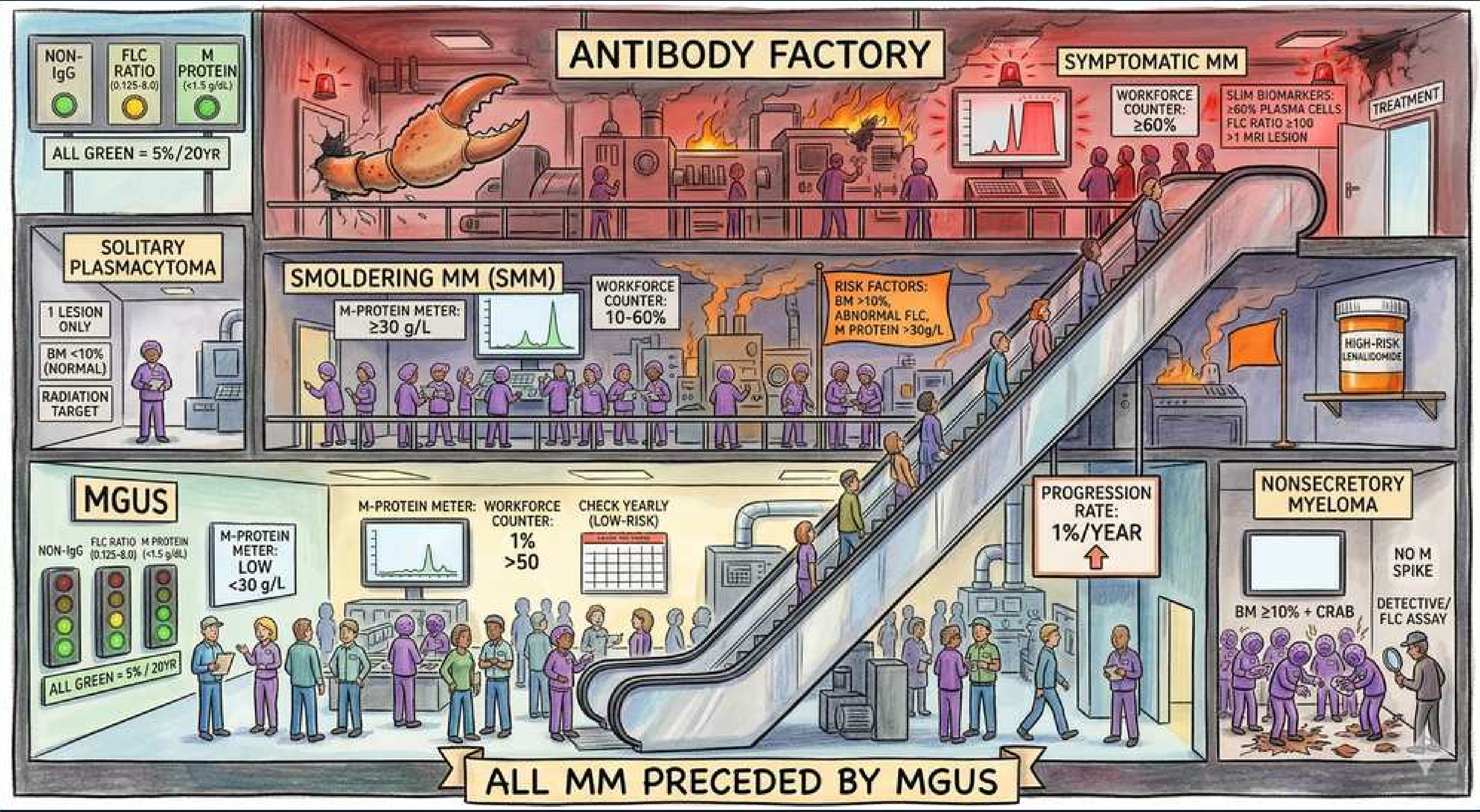
MGUS → Smoldering → Myeloma Spectrum — The Antibody Factory



1. MGUS — definition

WHAT YOU SEE
MGUS room: M-protein meter low, all green.

WHAT IT ENCODES
MGUS = M-protein <3 g/dL, clonal marrow plasma cells <10%, and NO CRAB end-organ damage. It is asymptomatic and premalignant.

BOARD TRAP
All three MGUS criteria must hold (<3 g/dL, <10% PCs, no CRAB). Any CRAB feature pushes the diagnosis to myeloma, not MGUS.

2. MGUS progression rate (~1%/yr)

WHAT YOU SEE
All-green traffic lights, ~1%/20yr placard.

WHAT IT ENCODES
MGUS progresses to myeloma/related malignancy at roughly 1% per year, and the risk persists lifelong, so indefinite monitoring is needed.

BOARD TRAP
MGUS risk is ~1%/year and does NOT disappear with time — lifelong follow-up, even years out.

3. Smoldering multiple myeloma (SMM)

WHAT YOU SEE
SMM hall: M-protein meter ≥30 g/L.

WHAT IT ENCODES
SMM = M-protein ≥3 g/dL OR clonal marrow plasma cells 10-60%, still WITHOUT CRAB or myeloma-defining events. Intermediate between MGUS and active myeloma.

BOARD TRAP
SMM has higher burden than MGUS (≥3 g/dL or 10-60% PCs) but still NO end-organ damage — the absence of CRAB is what keeps it 'smoldering'.

4. SMM progression rate (~10%/yr early)

WHAT YOU SEE
Workforce counter 10-60%.

WHAT IT ENCODES
SMM progresses ~10% per year in the first 5 years — much faster than MGUS — then the rate declines. High-risk SMM may warrant early therapy on trials.

BOARD TRAP
SMM ≠ MGUS: ~10%/yr early progression vs ~1%/yr. High-risk SMM is increasingly treated early, unlike MGUS.

5. Symptomatic (active) multiple myeloma

WHAT YOU SEE
Symptomatic MM bay: CRAB, ≥60% PCs, treatment.

WHAT IT ENCODES
Active MM = clonal plasma cells with CRAB (hyperCalcemia, Renal failure, Anemia, Bone lesions) OR a myeloma-defining biomarker. Requires treatment.

BOARD TRAP
It's the presence of CRAB or a myeloma-defining event — not just M-protein size — that defines treatable active myeloma.

6. Myeloma-defining biomarkers (SLiM)

WHAT YOU SEE
Sign: ≥60% PCs, FLC ratio ≥100, >1 MRI lesion.

WHAT IT ENCODES
Even without CRAB, treat if any 'SLiM' biomarker: clonal marrow PCs ≥60%, involved/uninvolved FLC ratio ≥100, or >1 focal lesion on MRI. These predict imminent progression.

BOARD TRAP
SLiM criteria upgrade SMM to active myeloma WITHOUT CRAB — ≥60% PCs, FLC ratio ≥100, or >1 MRI lesion all mandate treatment.

7. Solitary plasmacytoma

WHAT YOU SEE
Solitary plasmacytoma room: 1 lesion, BM <10%.

WHAT IT ENCODES
A single bone or soft-tissue plasma-cell tumor with normal marrow (<10% clonal PCs) and no CRAB. Treated with localized RADIATION; many later progress to MM.

BOARD TRAP
Solitary plasmacytoma is treated with radiotherapy, not systemic chemo — but monitor closely, as progression to MM is common.

8. Nonsecretory myeloma

WHAT YOU SEE
Nonsecretory room: no M-spike, detective FLC.

WHAT IT ENCODES
~1-2% of myeloma makes little/no detectable M-protein on SPEP/UPEP. Diagnosed by marrow plasmacytosis ≥10% + CRAB; the FLC assay often still detects involved light chains.

BOARD TRAP
A negative SPEP/UPEP does NOT exclude myeloma — nonsecretory disease is caught by FLC assay and marrow biopsy.

9. Risk factors for progression

WHAT YOU SEE
Risk-factor sign: abnormal FLC, M-protein ≥30 g/L.

WHAT IT ENCODES
Higher M-protein, abnormal FLC ratio, non-IgG isotype, and higher marrow PC% raise progression risk from MGUS/SMM to active disease and guide follow-up intensity.

BOARD TRAP
Risk stratification (M-protein size, FLC ratio, isotype, PC%) determines monitoring frequency — not all MGUS/SMM are equal.

10. All MM is preceded by MGUS

WHAT YOU SEE
Banner: 'All MM preceded by MGUS.'
WHAT IT ENCODES
Essentially all multiple myeloma evolves from a preceding MGUS phase, supporting the biologic continuum MGUS → SMM → MM.
BOARD TRAP
The continuum is one-directional: MGUS precedes virtually all myeloma, but most MGUS never progresses.

11. MGUS monitoring (check yearly)

WHAT YOU SEE
Check-yearly calendar, low-risk.
WHAT IT ENCODES
Low-risk MGUS is followed with periodic labs (SPEP, FLC, CBC, calcium, creatinine), typically annually, without treatment.
BOARD TRAP
MGUS is observed, not treated. Starting myeloma therapy for asymptomatic MGUS is a classic wrong answer.

12. SMM workforce 10-60% plasma cells

WHAT YOU SEE
Crowded SMM floor, counter 10-60%.
WHAT IT ENCODES
The 10-60% marrow plasma-cell range (without CRAB/SLiM) defines smoldering disease; ≥60% crosses into active myeloma by the SLiM criteria.
BOARD TRAP
60% is the cutoff: 10-60% PCs without CRAB = SMM, but ≥60% PCs = active myeloma regardless of symptoms.

13. High-risk SMM upgrade

WHAT YOU SEE
High-risk SMM upgraded, escalator up.
WHAT IT ENCODES
High-risk SMM (e.g., high M-protein + abnormal FLC + high PC%) has the greatest near-term progression risk; early lenalidomide-based therapy is being adopted to delay progression.
BOARD TRAP
High-risk SMM increasingly gets early treatment — a shift from the older 'always observe SMM' teaching.

14. M-protein meters across stages

WHAT YOU SEE
M-protein meters rising stage to stage.
WHAT IT ENCODES
Quantitative M-protein rises along the continuum (<3 g/dL in MGUS, ≥3 g/dL marks SMM/MM) and is the serial marker tracked across the spectrum.
BOARD TRAP
The same 3 g/dL serum M-protein line separates MGUS from SMM — memorize the numeric cutoffs that define each stage.

15. Symptomatic MM gets treatment

WHAT YOU SEE
Treatment door at the top floor.
WHAT IT ENCODES
Only active/symptomatic myeloma (CRAB or SLiM) is treated with systemic therapy; the earlier stages are observed (with high-risk SMM the evolving exception).
BOARD TRAP
Trigger to treat = CRAB or a myeloma-defining event, not the mere presence of an M-protein.